

Non-Parametric Spectral Estimation

Bartlett, Welch, and the resolution–variance trade-off

Learning objectives

ROADMAP

0:00–0:03

Explain how averaging periodograms reduces estimator variance.

Construct Bartlett and Welch PSD estimates from finite data.

Explain what segmentation, window choice, and overlap each change.

Reason quantitatively about resolution versus variance.

Convert Welch PSDs into stable features for classical ML.

Core question

How much frequency detail are we willing to trade for a PSD estimate that is statistically more stable?

The central trick: average noisy estimates

BARTLETT

0:03–0:08

One periodogram:

high frequency detail + high variance

Average K approximately independent periodograms:

variance roughly decreases like $1/K$

But where do K estimates come from?

Split one long record into shorter segments. That gains averaging at the cost of shorter observation length per estimate.

Bartlett's method

BARTLETT

0:08–0:15

1. Split N samples into K non-overlapping segments of length $L = N/K$.
2. Compute a periodogram $I_k(f)$ for each segment.
3. Average them:

$$\hat{S}_B(f) = (1/K) \sum_{k=0}^{K-1} I_k(f)$$

Trade-off

L smaller $\rightarrow$ more segments $K \rightarrow$ lower variance, but coarser frequency resolution.

Bartlett: count the arithmetic and the information

BARTLETT

0:15–0:20

Example: $N = 4096$ samples

$K = 8 \Rightarrow L = 512$ samples per segment

At $F_s = 1000$ Hz:

nominal bin spacing = $F_s/L \approx 1.95$ Hz

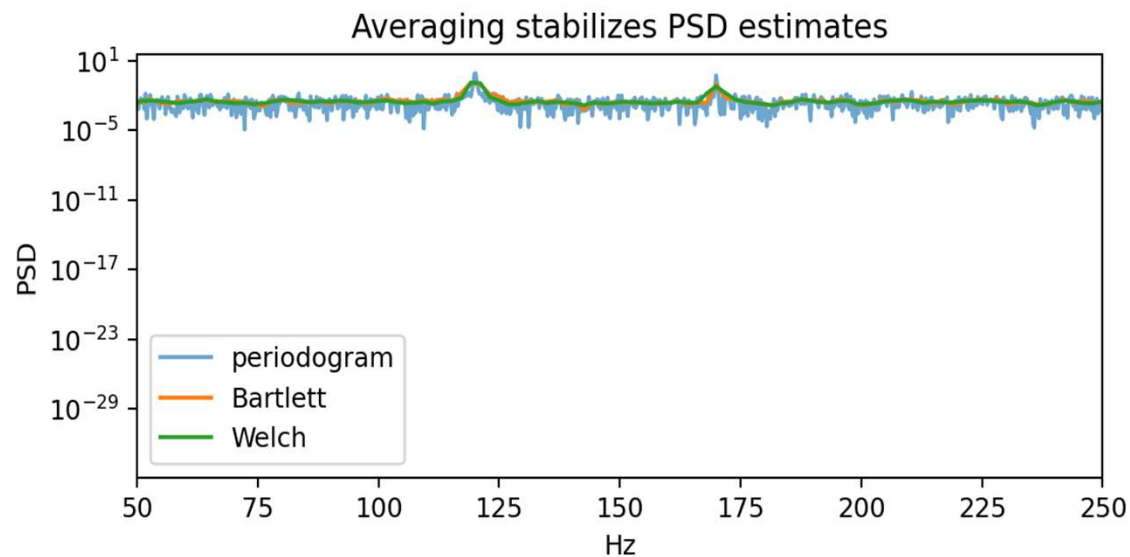
Compare with full record

The 4096-sample periodogram has ~ 0.244 Hz bin spacing. Bartlett gives up that fine grid to obtain 8 estimates to average.

Periodogram vs. Bartlett vs. Welch

METHOD COMPARISON

0:20–0:24



What changes?

Periodogram is detailed but jagged. Bartlett is smoother. Welch adds tapering and overlap, typically yielding a stable practical estimate.

Do not interpret “smoother” as automatically “more accurate” at every frequency.

Why Welch improves on Bartlett

WELCH

0:24–0:30

Use overlapping segments instead of disjoint segments.

Apply a tapering window to each segment before the FFT.

Normalize for window power so the PSD scale remains meaningful.

Average the modified periodograms.

Two goals

Windowing controls leakage.
Overlap recovers more averaging opportunities after tapering discards weight near segment edges.

Welch's estimator: algorithm view

WELCH

0:30–0:36

For segment k with window $w[n]$:

$$X_k(f) = \sum_n w[n] x_k[n] e^{-j2\pi fn/F_s}$$

Modified periodogram:

$$I_k(f) \propto |X_k(f)|^2 / \sum_n w^2[n]$$

Then average $I_k(f)$ across segments.

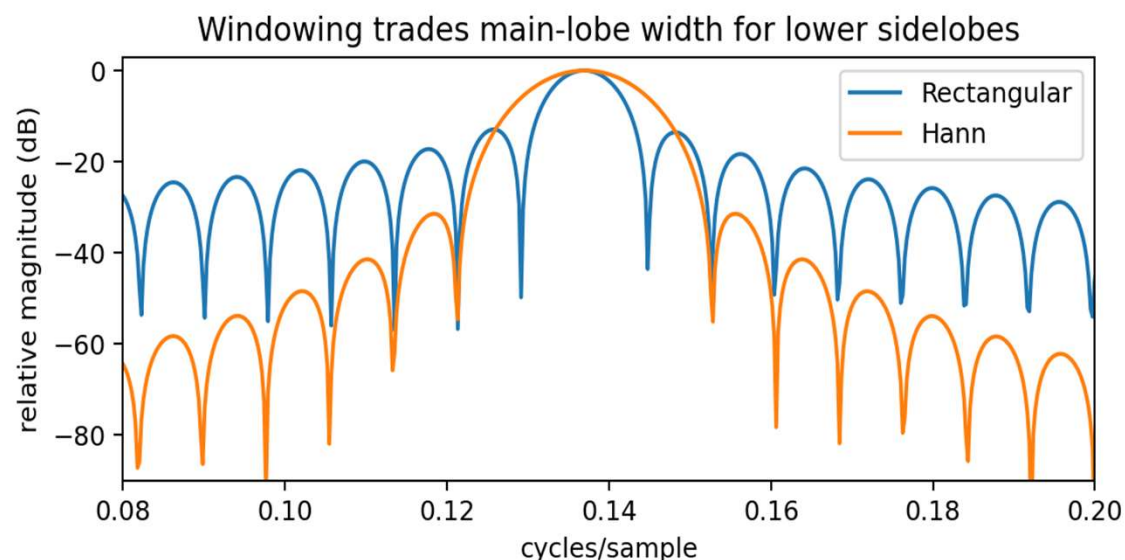
Implementation

`scipy.signal.welch` handles window normalization, one-sided scaling, overlap, and FFT details.

Window choice: leakage versus resolution

WELCH

0:36–0:42



Rectangular

Narrower main lobe, but high sidelobes. Good resolution for bin-centered tones; poor leakage behavior otherwise.

Hann / Hamming

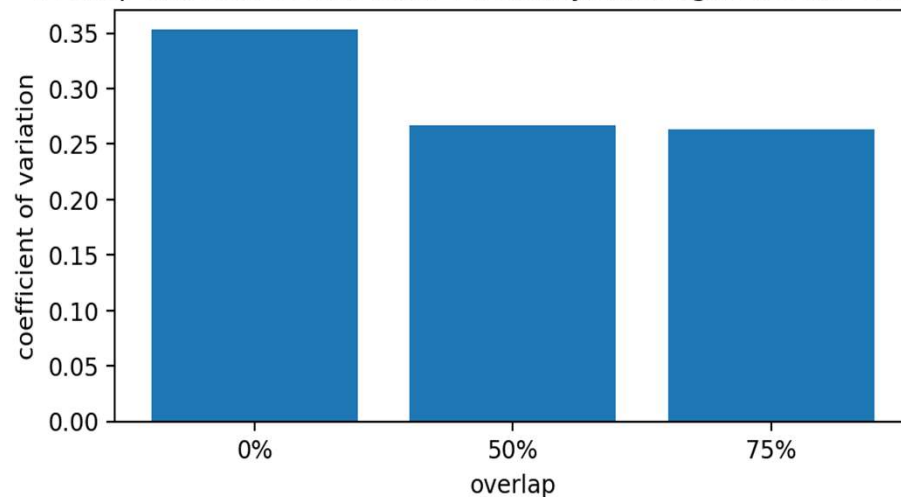
Lower sidelobes and better leakage control, but wider main lobes. Nearby spectral features can blur together.

Overlap: more segments, not more independent data

WELCH

0:42–0:47

Overlap can reduce estimator variability, but segments are correlated



0% overlap: fewer, more nearly independent segments.

50% overlap: common with Hann; reuses samples while shifting the window support.

75% overlap: even more estimates, but substantial correlation between neighboring segments.

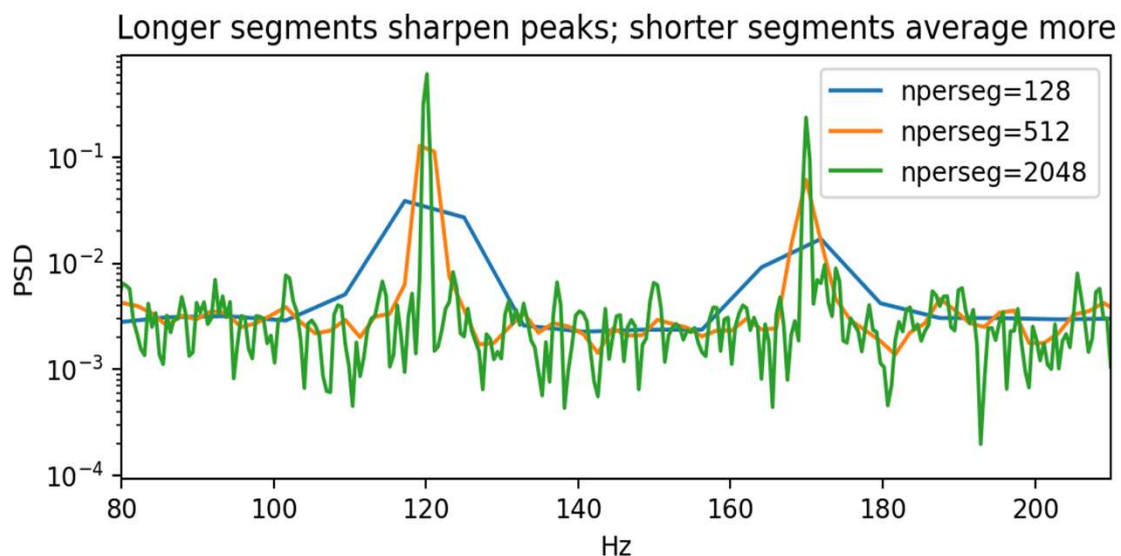
Key nuance

Variance reduction is not exactly $1/K$ when segments overlap because the estimates are correlated.

Segment length controls the main trade-off

RESOLUTION vs VARIANCE

0:47–0:53



Long segments

Sharper peaks and finer frequency detail, but fewer averages $\rightarrow$ higher estimator variance.

Short segments

More averages and smoother PSD, but broad peaks and reduced ability to separate nearby components.

A quantitative design recipe

RESOLUTION vs VARIANCE

0:53–0:58

Start from the frequency detail you need.

Choose segment length L so F_s/L is comfortably smaller than the separation of important features.

Then choose a window and overlap.

Window: leakage suppression versus main-lobe width.

Overlap: number of averaged segments versus correlation/computation.

Validate on repeated records, not just one aesthetically smooth PSD.

Rule of thumb, not law

50% overlap with a Hann window is a strong default, but the signal and objective should determine the final design.

Concept check: two close tones in noise

CONCEPT CHECK

0:58–1:02

You must distinguish tones at 120 Hz and 126 Hz in noisy data sampled at 1 kHz.

Which change is most likely to help?

- A. Shorter Welch segments
- B. Longer Welch segments
- C. More zero-padding only
- D. Heavier smoothing across frequency

Good answer: B

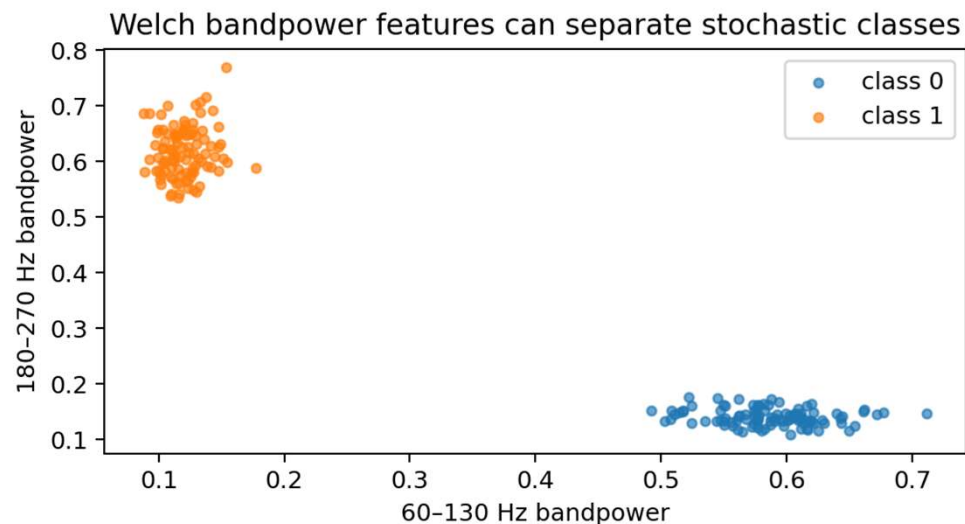
Longer segments provide narrower spectral main lobes and better true frequency resolution. Zero-padding alone only samples the same spectral estimate more densely.

PSD as an ML feature map

ML BRIDGE

1:02–1:07

Raw waveform → Welch PSD → band powers / spectral shape → classifier



Welch features average away nuisance sample-to-sample randomness.

Bandpower compresses hundreds of PSD bins into interpretable features.

StandardScaler + SVM or Random Forest can then operate on stable vectors.

Caution

PSD features intentionally discard phase and precise time localization. That may be desirable—or fatal—depending on the task.

Example ML pipeline

ML BRIDGE

1:07–1:11

For each record:

1. Compute Welch PSD with a fixed analysis configuration.
2. Integrate power in physically meaningful frequency bands.
3. Optionally add spectral centroid, roll-off, or log-band powers.
4. Standardize features using training data only.
5. Train SVM / logistic regression / random forest and cross-validate.

Leakage in ML can be literal

Never compute normalization statistics using the test set. Signal-processing leakage and data-set leakage are different problems—but both invalidate conclusions.

Coding demo: design Welch instead of just calling it

DEMO

1:11–1:14

Compare custom Bartlett and `scipy.signal.welch`.

Sweep `nperseg` and observe peak separation versus smoothness.

Sweep overlap and measure repeated-estimate variability.

Build Welch bandpower features and train an SVM/Random Forest.

Student prediction first

Before running each cell, ask which curve should be smoother, which should resolve the tones better, and why.

The goal is estimator reasoning—not memorizing one “best” Welch configuration.

Lecture 18 wrap-up

WRAP-UP

1:14–1:15

Bartlett reduces variance by averaging non-overlapping periodograms.

Welch adds windows and overlap for better leakage control and practical averaging.

Segment length is the primary resolution–variance knob.

Overlap helps, but overlapping segments are correlated.

Stable PSD summaries can be strong classical-ML features.

Exit ticket

If a Welch PSD looks too noisy, name one change that reduces variance and state the price you pay.